

RA8P1 E-READER

Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

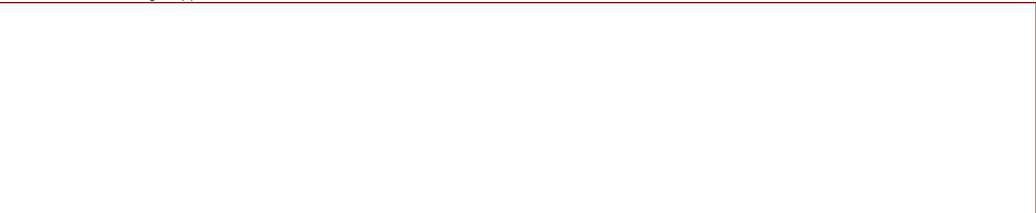
RABP1 IO allocation



RABP1 internal core regulator



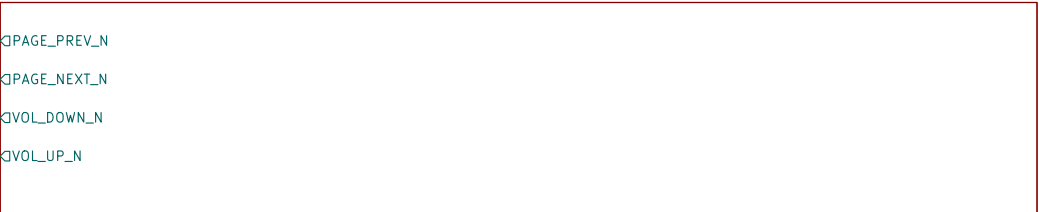
RABP1 IO and analog supplies



RABP1 clocks, reset and debug



Page and volume buttons



ESP32-C6 radio

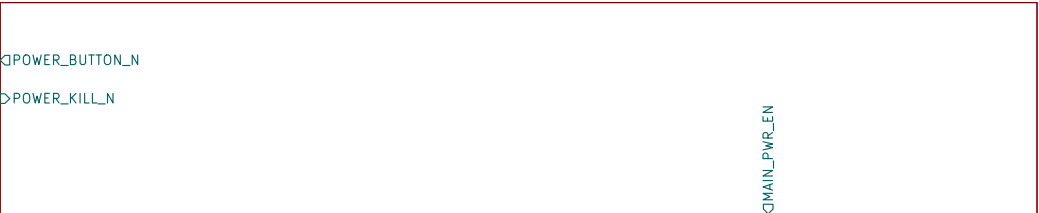


Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

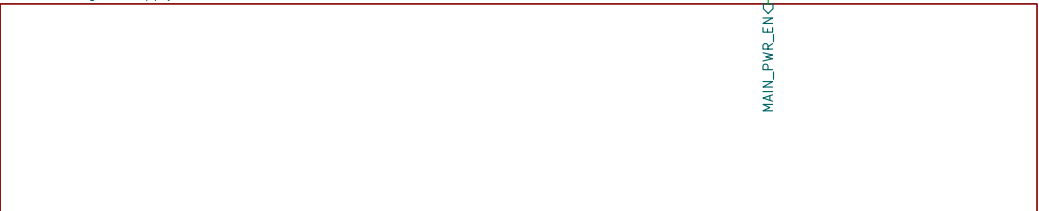
Remaining subsystem circuits follow epic #821 and issues #824–#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

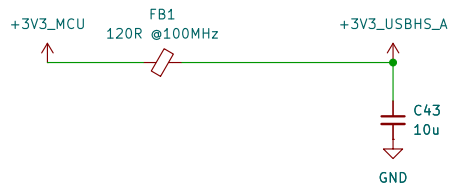
Power button and source control



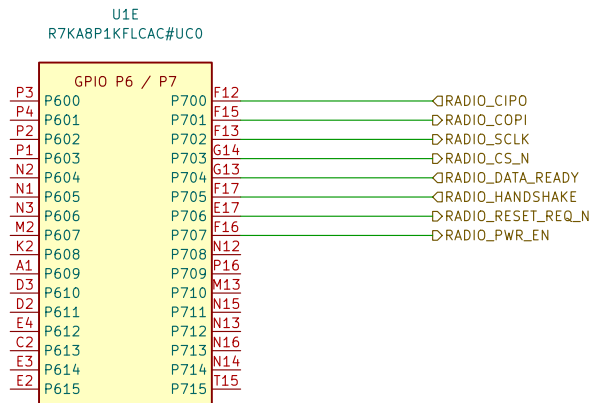
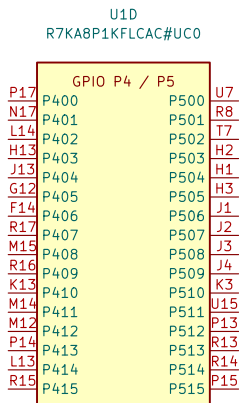
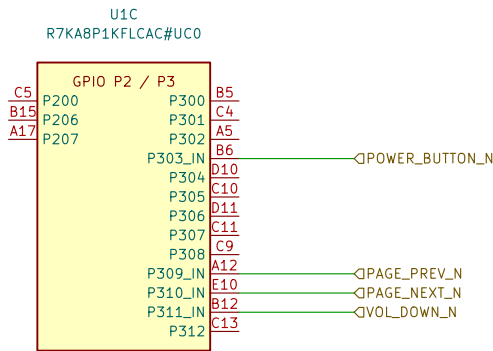
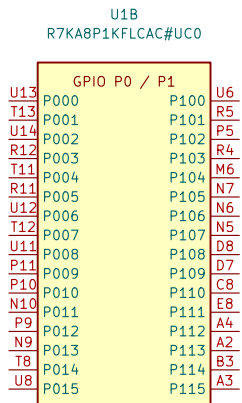
Main 3V3 digital supply



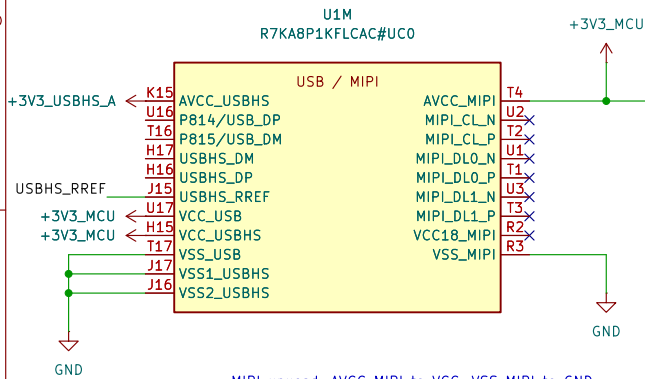
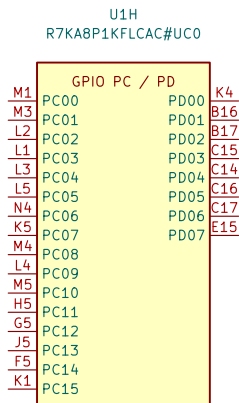
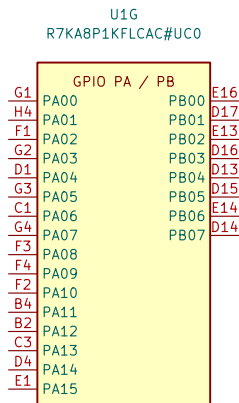
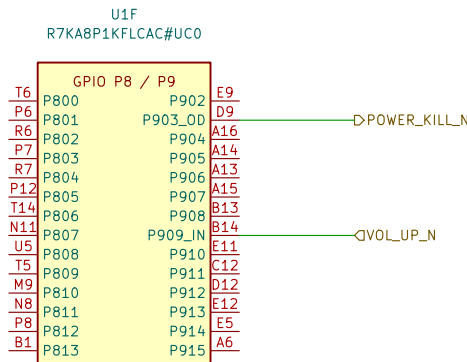
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



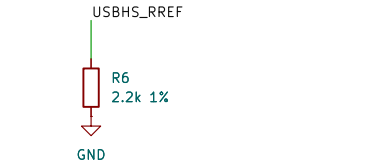
USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm; 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9.11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification; verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]



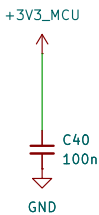
RADIO-008: SPIA_C P700..P703; IRQ26=P704, IRQ19=P705.
P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals.
Switched-domain isolation and reset arbitration remain open.
Pin map + conflicts: ../design/radio_interface.md#radio-008-ra8p1-host-allocation



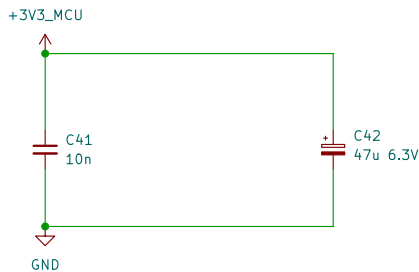
MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.



USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md



USB-002: USB full-speed supply / U1,U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90..110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md



USB-003: USB-HS supply / U1,H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1 +/-0.10) = 9..11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1 +/-0.20) = 37.6..56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90..110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

RA8P1 e-reader

Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

Title: RA8P1 IO allocation

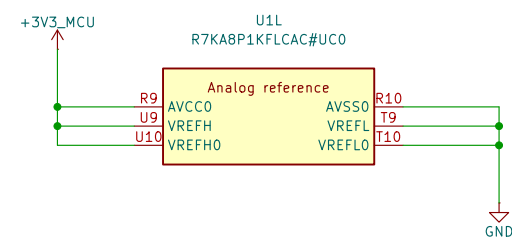
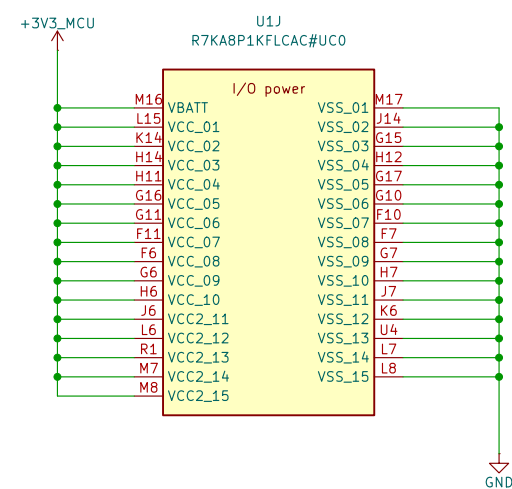
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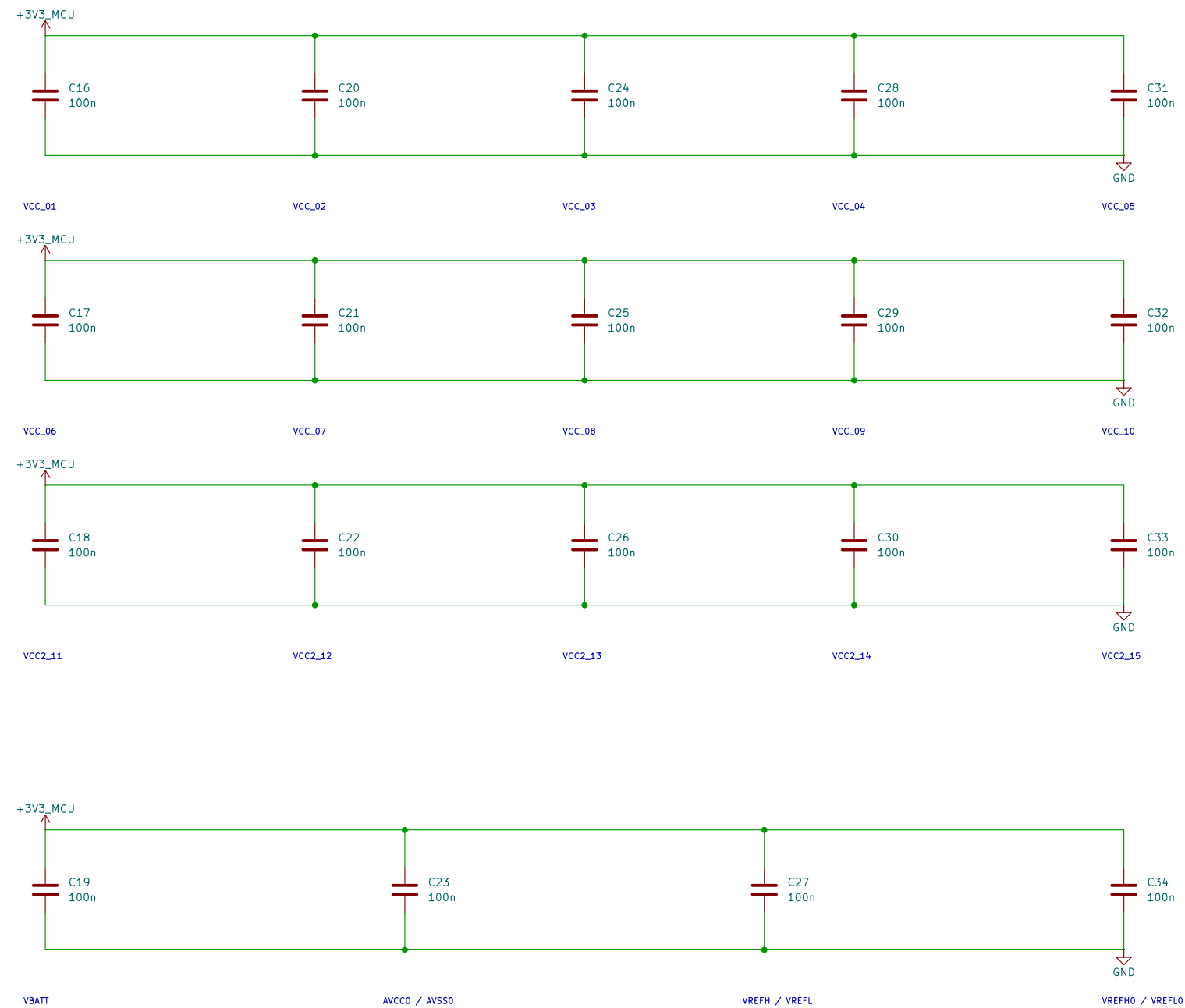
Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.5

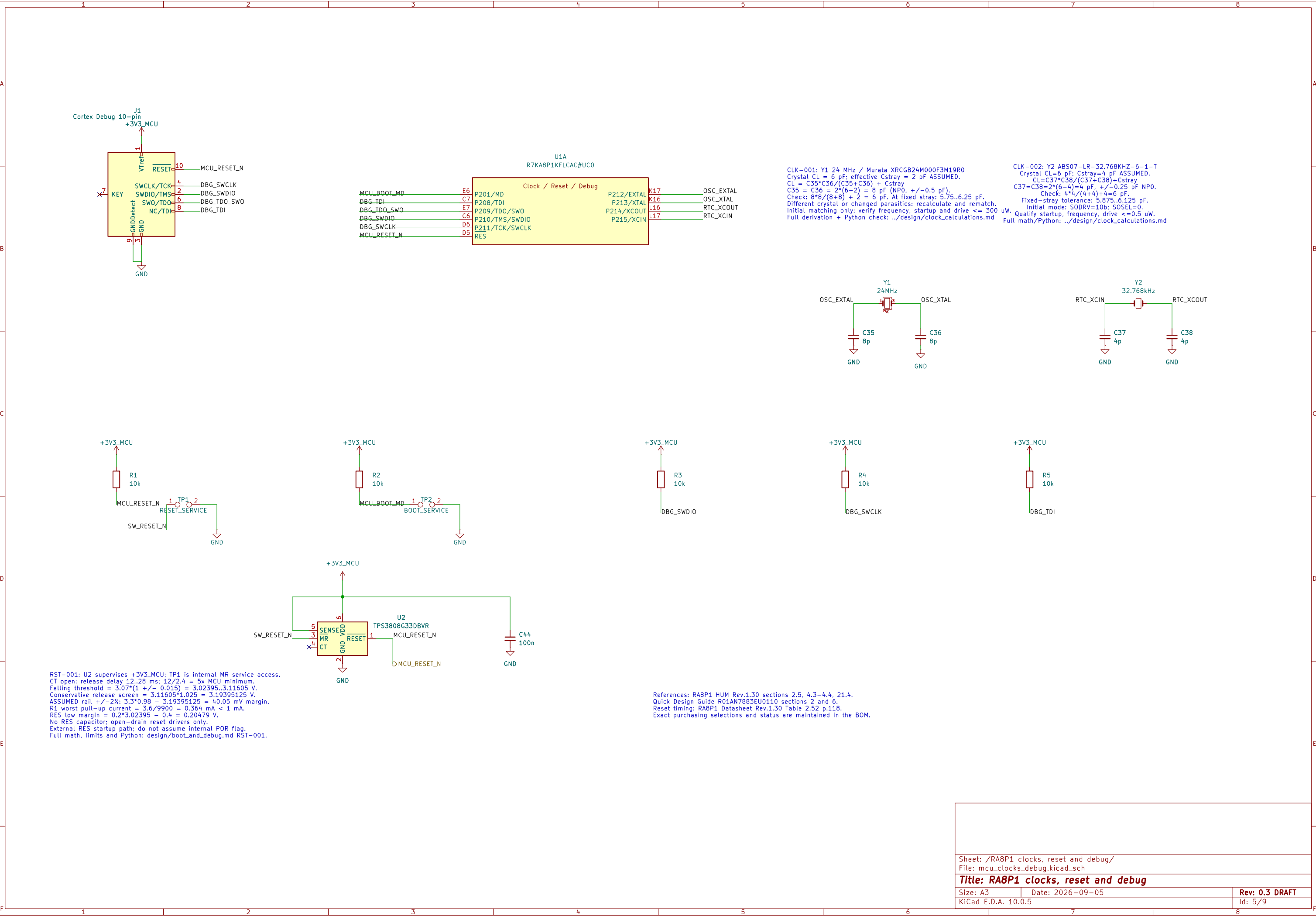
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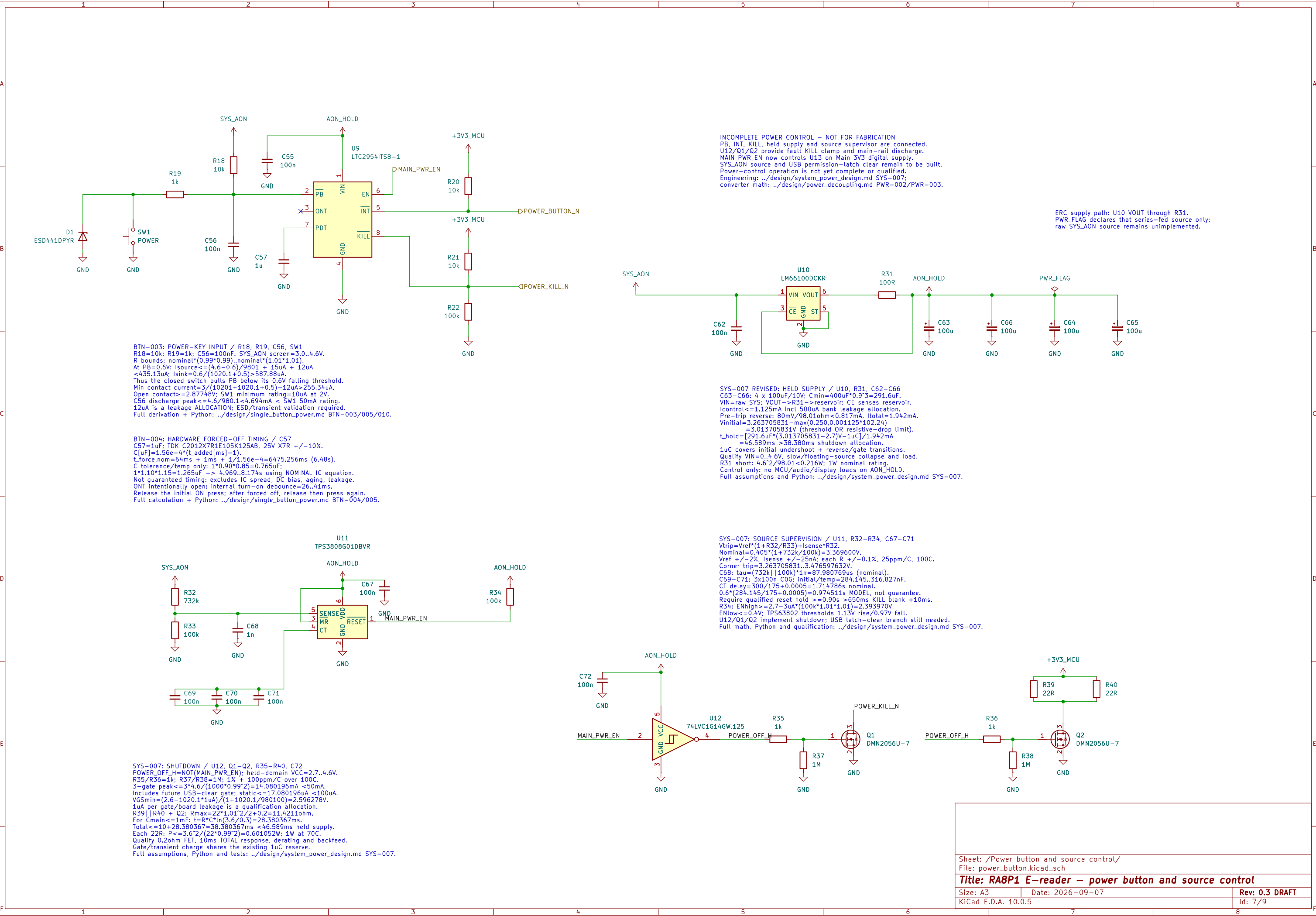


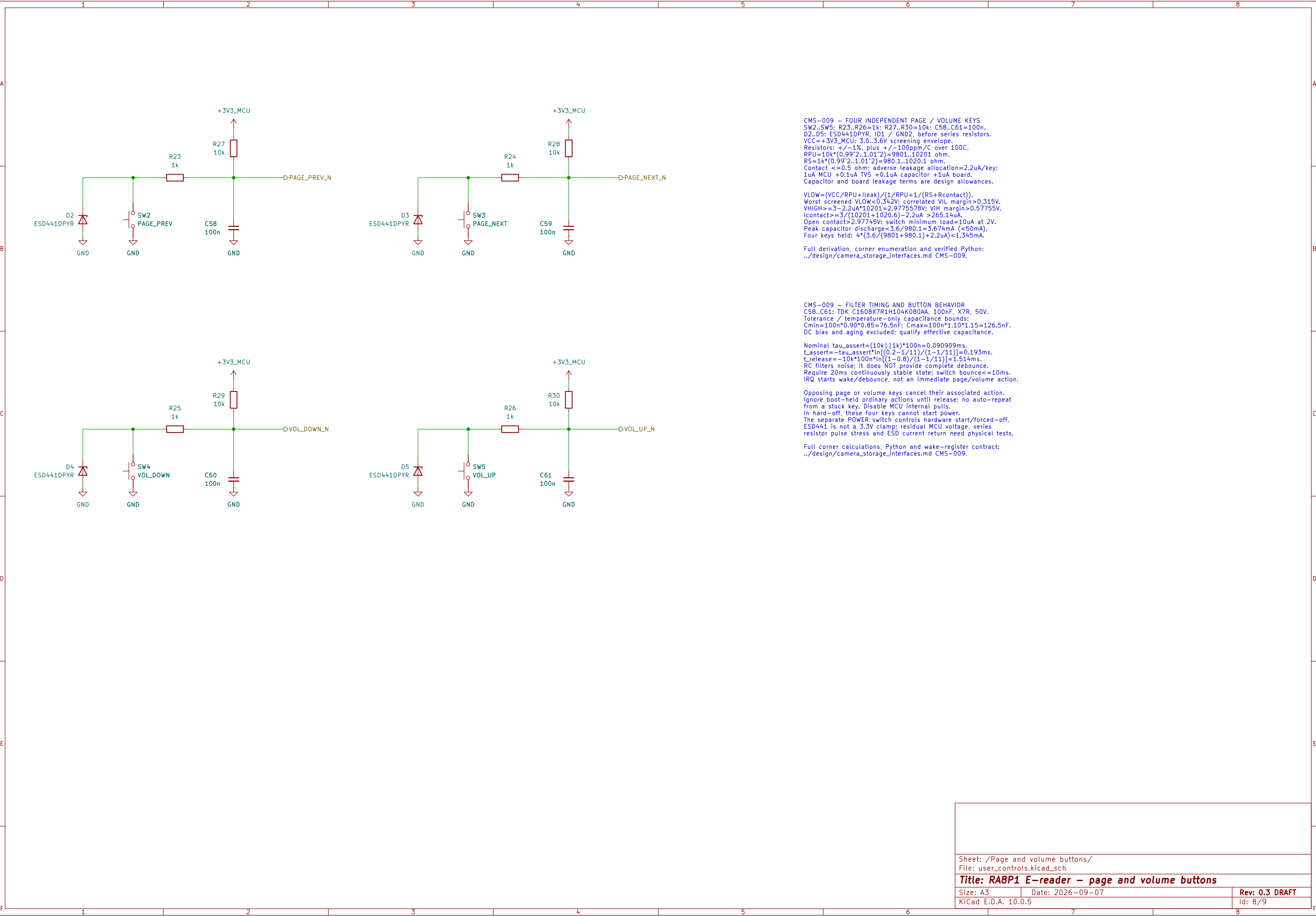
I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR



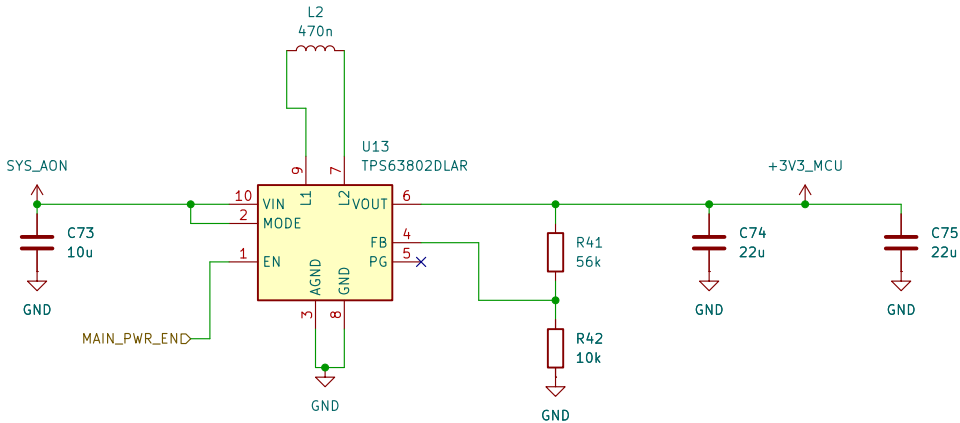
VCC and VCC2 use 3.3 V. All returns use one board ground plane.
VBATT follows the main rail; no separate backup-cell supply.
Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs.
USB/MIP1 supplies remain on the interface sheet and are not yet wired.
HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862.
This is an electrical draft; capacitor MPN qualification remains open.







MAIN DIGITAL 3.3 V SUPPLY
TPS63802 buck–boost; hardware–controlled enable; fixed PWM.
Engineering draft: upstream source and corner qualification remain open.



PWR–002 / PWR–003: SETPOINT AND LOCAL COMPONENTS
U13 TPS63802DLAR; MODE=VIN, fixed PWM.
R41=56k; R42=10k; each +/-0.1%, 25ppm/C.
Vnom=0.500*(1+56k/10k)=3.300V.
Idivider=3.300/(56k+10k)=50uA.
For 100C resistor–temperature change:
fLo=0.999*0.9975=0.9965025; fHi=1.001*1.0025=1.0035025.
Vmin=0.495*(1+56k*fLo/(10k*fHi))–100nA*56k*fHi
=3.242044V.
Vmax=0.505*(1+56k*fHi/(10k*fLo))+100nA*56k*fHi
=3.358485V.
Static screen only; ripple, load steps and drift still require validation.
C73: CIN_model=10u*0.90*0.85*0.60=4.590uF >=4uF.
C74+C75: COUT_model=2*22u*0.90*0.85*0.60=20.196uF >=7uF.
0.90=tolerance; 0.85=X7R temperature; 0.60=residual ALLOCATION.
Qualify bias, aging, excitation and assembly; 60% is not guaranteed.
L2: 0.47uH +/-20% gives 0.376..0.564uH INITIAL only.
Required installed effective L=0.37..0.57uH; bias/reflow/hot tests OPEN.
TI SLVSEU9D Table 7–1, sections 9.4/10.2; exact parts in native BOM.
Full derivation, primary sources and executable Python:
../design/power_decoupling.md PWR–002/PWR–003.

PWR–003: LOAD, SOURCE AND STARTUP ACCEPTANCE LIMITS
Main continuous allocation: 0.750 MCU + 0.500 radio
+ 0.250 SDRAM + 0.100 NOR + 0.050 logic = 1.650A.
These are qualification allocations, not measured maximum currents.
SD, camera, audio, display/touch/frontlight need separate SYS supplies.
Pout=3.358485094*1.65=5.541500W.
VIN>=3.20V allocation; raw trip min=3.263705831V.
Allowed raw–to–VIN loss=3.263705831–3.20=63.705831mV.
At ASSUMED eta=85%: Iin=Pout/(eta*3.20)=2.037316A.
Ploss=Pout*(1/eta–1)=0.977912W.
Reference rise=Ploss*81C/W=79.210859C; NOT enclosure prediction.
Verify hot current capability and junction temperature <125C with margin.
Cold subtotal, RADIO OFF: 1.330 DCDC + 0.400 other = 1.730A.
Radio on would give 2.230A > nominal 2A; keep radio OFF through start/wake.
Largest wake–reference subtotal=1.270+0.400=1.670A.
DCDC figures are reference–only; non–DCDC start and extra charging excluded.
Additional charging: I=Cadditional*dV/dt; qualify the actual ramp.
Total directly connected main–rail C <=1mF for SYS–007 discharge budget.
No externally driven +3V3_MCU: forced PWM can sink reverse current.
PWR–002 static release margin: radio 14.922537mV; MCU 48.092861mV.
Ripple, startup and transients must fit; no fabrication release.
Full sources, derivation and Python: ../design/power_decoupling.md PWR–003;
source–loss shutdown: ../design/system_power_design.md SYS–007.